

10(a)	<i>any seven from:</i> 1 action potential / depolarisation / impulse, at pre-synaptic <u>membrane</u> ; 2 (voltage-gated) calcium ion channels open / calcium ions enter (cell / cytoplasm / (motor) neurone / pre-synaptic knob) ; 3 vesicles fuse with pre-synaptic membrane ; 4 acetylcholine / ACh, released, by exocytosis / into synaptic cleft ; 5 (ACh) binds to receptors on, muscle cell membrane / sarcolemma / motor end plate ; 6 sodium ion channels open / sodium ions enter (muscle cell / sarcoplasm) ; 7 depolarisation of, (muscle) cell surface membrane / sarcolemma ; 8 (depolarisation) spreads / transmitted, to / down / via, T-tubules ; 9 depolarisation of (adjacent) sarcoplasmic reticulum (membrane) ; 10 (voltage-gated) calcium ion channels open ; 11 calcium ions, move / diffuse, out of SR / out of cisterna(e) ; 12 calcium ions, move / diffuse, into, sarcoplasm / cytoplasm ; 13 calcium ions, start contraction / bind to troponin ;	7
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Question	Answer	Marks
10(b)	<i>any eight from:</i> 1 mutation / allele / gene, on chromosome 4 / autosome ; 2 dominant ; 3 normal / recessive, allele has 10–35 repeats of CAG ; 4 HD / dominant / mutant, allele has, more / extra, repeats of CAG ; 5 larger number of repeats gives earlier onset ; 6 usual onset, after 28 / in middle age / before 65 ; 7 onset, in babies / from 1 year old, if very numerous repeats ; 8 ref. extra glutamine / polyglutamine ; 9 mis-folded, protein / huntingtin ; 10 neurological condition / brain problem OR (brain) neurones, die / destroyed ; 11 motor control uninhibited / involuntary movements / chorea ; 12 cognitive / mood, changes ; 13 AVP ; e.g. GABA producing neurones lost <i>ref. to basal ganglia / striatum, affected (first)</i>	8